SDG BLOCKCHAIN ACCELERATOR

Impact Measurement Framework

1. Project Information

Project Name
ClimaFi – Blockchain-Based Climate Impact Verification and Tokenization Platform

Challenge & UNDP Office
Climate finance and impact initiatives often suffer from fragmented Measurement, Reporting, and Verification (MRV) systems, lack of transparency, and limited trust between stakeholders. This results in inefficient capital allocation, difficulty in verifying impact claims, and barriers to scaling climate-positive projects.
Challenge Owner: Better Than Cash Alliance / UNDP

2. Measurement Objective and Approach

This framework is designed to assess whether ClimaFi successfully establishes a **credible, verifiable, and scalable infrastructure for climate impact measurement and monetization**.

The evaluation focuses on three core dimensions:

- **Impact Integrity:** Accuracy, auditability, and reliability of climate data (MRV)
- **System Adoption:** Uptake by projects, institutions, and ecosystem actors
- **Economic Activation:** Ability to unlock funding and incentives through tokenized impact

The framework follows a **results-based approach**, linking on-chain verification to real-world environmental and financial outcomes.

3. Theory of Change

ClimaFi introduces a blockchain-based layer that anchors climate impact data (e.g., carbon, energy, environmental metrics) on-chain and enables tokenization of verified outcomes.

This enables:

- Standardized and tamper-proof recording of climate data
- Transparent and auditable impact verification
- Creation of digital assets representing verified impact

Which leads to:

- Increased trust between project developers, funders, and regulators

- Reduced verification costs and delays
- Improved access to climate finance mechanisms

Ultimately contributing to:

- Greater capital flow into climate-positive projects
- Higher integrity in impact reporting
- Scalable infrastructure for global climate finance systems

4. SDG Alignment and Indicators

The project contributes directly to:

SDG 13 – Climate Action

- Verified climate impact data supporting mitigation efforts
- Indicator: Volume of CO_2 (or equivalent) verified and recorded

SDG 9 – Industry, Innovation and Infrastructure

- Deployment of digital infrastructure for MRV systems
- Indicator: Number of projects and institutions using the platform

SDG 17 – Partnerships for the Goals

- Multi-stakeholder coordination through shared data infrastructure
- Indicator: Number of ecosystem actors integrated (developers, verifiers, funders)

5. Key Performance Indicators (KPIs)

5.1 System Performance (Infrastructure Validity)

Indicator	Baseline	Target	Data Source	Frequency
System uptime	N/A	≥99%	System logs	Monthly
Data anchoring success rate	0%	≥98%	Blockchain logs	Monthly
Avg verification processing time	High / manual	-50% reduction	Platform logs	Monthly

Transaction success rate	N/A	≥98%	Blockchain logs	Monthly

5.2 Impact Integrity (MRV Credibility)

Indicator	Baseline	Target
% of impact data verifiable on-chain	0%	100%
% of datasets with audit trail	0%	100%
Data discrepancy rate (audit vs reported)	Unknown	<5%
Verification cycle duration	High	-50% reduction

5.3 Adoption & Ecosystem Engagement

Indicator	Baseline	Target
Number of projects onboarded	0	≥3 (pilot)
Number of institutional partners	0	≥3
Active users (monthly)	0	≥70% of onboarded
Number of verified impact events recorded	0	≥100+

5.4 Economic Activation (Climate Finance Enablement)

Indicator	Baseline	Target
Value of tokenized impact assets	0	Pilot issuance completed
Number of impact tokens issued	0	≥1 issuance cycle
Time from verification → monetization	High	-50% reduction

Number of funding interactions enabled	0	≥3

5.5 Transparency & Trust

Indicator	Baseline	Target
% of stakeholders able to independently verify data	0%	≥80%
Reported trust in impact data	Low	≥80% satisfaction
Number of disputes over data validity	Unknown	-70% reduction

6. Tracking Methods

The framework uses a hybrid **on-chain + off-chain data architecture**:

On-chain Metrics (Source of Truth)

- Impact data hashes and records
- Token issuance and lifecycle events
- Timestamped verification logs

Off-chain Metrics (Contextual Data)

- Project-level datasets (MRV inputs)
- Institutional engagement records
- User activity and onboarding data
- Surveys (trust, usability, adoption)

Data Validation Approach

- Cross-verification between submitted datasets and on-chain records
- Third-party or verifier validation (where applicable)
- Periodic reconciliation and audit sampling

7. Baseline vs Projected Outcomes

Baseline

- Fragmented and non-standardized MRV systems
- Limited transparency in impact reporting
- High verification costs and delays
- Low trust between stakeholders

Projected (Pilot Phase)

- Fully traceable and verifiable impact datasets
- Operational tokenization of climate impact
- Reduced verification time and increased efficiency
- Improved trust and transparency across stakeholders

Post-Pilot (Expected)

- Integration into broader climate finance workflows
- Expansion to additional projects and geographies
- Standardization of blockchain-based MRV approaches

8. Monitoring & Reporting Plan

Frequency

- Baseline assessment (pre-pilot)
- Monthly monitoring
- End-line evaluation

Responsible Parties

- ClimaFi (platform + data infrastructure)
- UNDP / Better Than Cash Alliance (oversight and validation)

Reporting Outputs

- Real-time dashboard (impact, usage, tokens)
- Monthly performance briefs (KPIs vs targets)
- Final evaluation report (impact, lessons, scale readiness)

9. Success Criteria

The pilot will be considered successful if it demonstrates:

- ≥100% traceability of recorded impact data

- ≥50% reduction in verification time
- Successful issuance of tokenized impact assets
- ≥3 projects actively using the system
- ≥80% stakeholder trust in data integrity

10. Risks & Mitigation

Data Quality Risk

- Poor or inconsistent input data
- *Mitigation:* Standardized data templates, validation layers

Low Adoption

- Limited engagement from projects or institutions
- *Mitigation:* Strong onboarding, aligned incentives

Overengineering / Complexity

- System too complex for users
- *Mitigation:* Simplified UX, modular design

Verification Bottlenecks

- Delays in validation processes
- *Mitigation:* Streamlined workflows, partial automation

Regulatory Uncertainty

- Ambiguity around tokenized assets
- *Mitigation:* Positioning as data/infrastructure layer

11. Scalability Considerations

The framework is designed to support scaling by:

- Standardizing impact data structures
- Enabling interoperability with existing MRV systems
- Supporting multiple asset types (carbon, energy, biodiversity)
- Providing a replicable model for UNDP country offices